HW 7.4

Principles of Microeconomics

October 11, 2025

1. Suppose the market demand for burritos is given by $Q_D = 40 - 5P$, and the market supply for burritos is given by $Q_S = 10P - 20$.

   (a) What are the equilibrium price and quantity?

   (b) Calculate consumer surplus and producer surplus. Hint: Draw a graph.

2. Demand for gasoline is given by $Q_D = 105 - 15P$. Supply is $Q_S = 25P - 15$.

   (a) What are price and quantity at the market equilibrium?

   (b) What are consumer and producer surplus at the market equilibrium?

   (c) To reduce greenhouse gas emissions, Congress passes a tax on gasoline. By law, producers have to pay \$1.60 in taxes for every gallon sold. What is the new quantity of gasoline sold?

   (d) What price do consumers pay, and what price do producers receive after the tax?

   (e) What are consumer and producer surplus at the market equilibrium?

   (f) What's government revenue after the tax?

   (g) What's deadweight loss after the tax?